4 **(a)** An experiment is performed using two large vertical plates maintained at potential + 6.0 V and – 6.0 V and separated by distance *d* as shown in Fig. 4.1.

When a particle of mass 0.25 g and charge *Q* is released from rest between the plates, it moves in a straight line at an angle θ of 45° to the vertical.

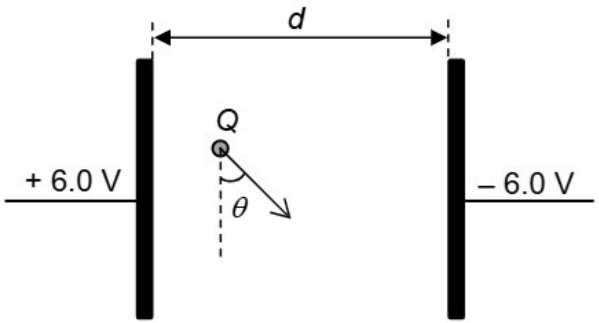


Fig. 4.1

(i) State and explain the sign of the charge on the particle.

.....

..... [1]

(ii) Determine the magnitude of the electric force acting on the charged particle.

electric force = ... N [1]

(b) The distance d between the plates is changed and the experiment is repeated with the particle being released from rest at the same initial position.

The graph of $\tan\theta$ against $\frac{1}{d}$ is shown in Fig. 4.2.

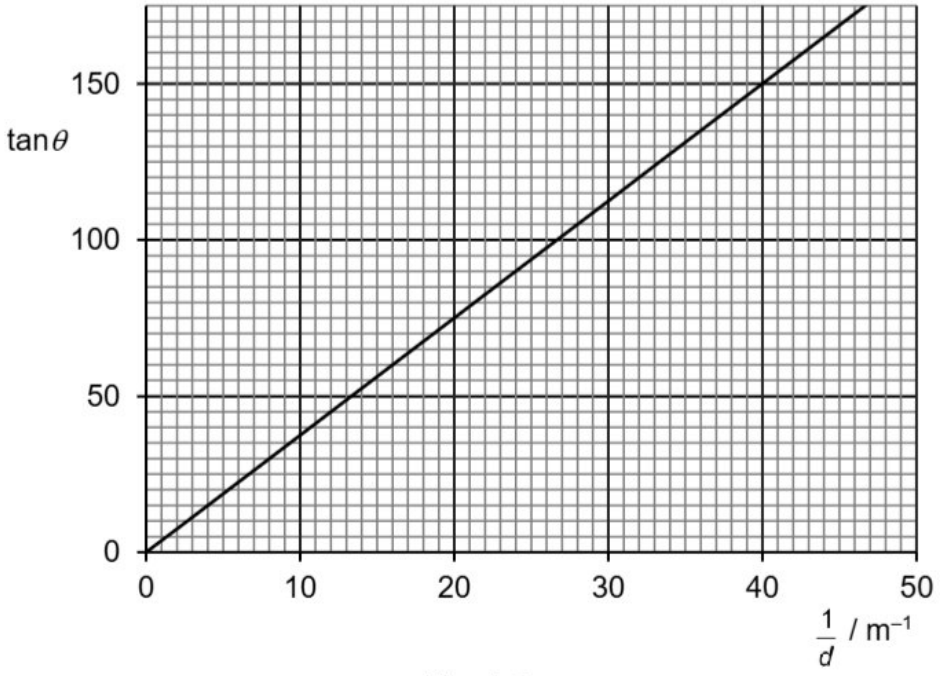


Fig. 4.2

Determine the magnitude of charge Q.

Q = ... C [3]

(c) The two large vertical plates in **(a)** form a parallel plate capacitor of capacitance 5.0 pF that is charged to 12.0 V.

This capacitor is then connected to another capacitor with capacitance of 10.0 pF that is initially charged to 6.0 V as shown in Fig. 4.3.

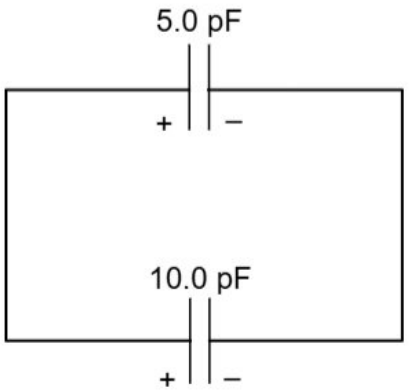


Fig. 4.3

Determine the final charge on each capacitor.

charge on the 5.0 pF capacitor = ... C

charge on the 10.0 pF capacitor = ... C [3]

[Total: 8]